MILITARY SPECIFICATION

TUBES AND TAPERS, PERISCOPE

1. SCOPE

1.1 This specification covers periscope outer tubes and tapers for Naval submarine periscopes.

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids, forms a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipments and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - The material shall be corrosion-resisting steel made by the electric furnace process such as to produce tubes and tapers in full compliance with the chemical and physical requirements of this specification.

3.1.1 Chemical composition. - The material shall conform to the chemical requirements specified in table I.

Table I - Chemical composition. 1/

Chromium (Min.)	Nickel (Min.)	Carbon (Max.)	Manganese (Max.)	Phosphorus (Max.)	Sulphur (Max.)	Silicon (Max.)	Copper (Max.)	Molybdenum (Max.)
Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
18.00-20.00	8.00-12.00	0.08	2.00	0.04	0.030	1.00	0.50	0.60

1/ Unless otherwise specified in the contract or order, elements other than those shown in table I shall not be present in excess of 0.50 percent for the sum of all such elements without the approval of the bureau or agency concerned. The presence of any single other element in excess of 0.05 percent shall be approved by the bureau or agency concerned.

3.1.2 Physical properties. - The material shall conform to the physical requirements specified in table II.

3.2 Design. - Tubes and tapers shall be in accordance with the design specified on the applicable drawings (see 6.1).

3.3 Construction. - Tubes and tapers shall be constructed by forging and boring or by piercing and drawing. Tubes and tapers shall each be of one piece, seamless and completely machined, both inside and outside. Where the tubes and the tapers are bored from the solid, the forging shall be of the smallest possible diameter consistent with final machining to size in order to insure thorough working of the metal which forms the wall of the completed tube.

3.4 General requirements. -

3.4.1 Dimensions. - The dimensions of the tubes and tapers shall be in accordance with the applicable drawings (see 6.1).

3.4.2 Straightness. -

3.4.2.1 Each tube and taper shall be a true and straight right circular section and of uniform diameter except for grooves, collars, and threads required by the applicable drawings. The bore of each shall be straight, of circular section, and of uniform diameter, except for grooves, collars or threads as required by the applicable drawings. The bore shall be parallel to, and concentric with, the outer surface.

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Table II - Physical properties.

Modulus of elasticity psi (Min.)	Proof stress psi (Min.)	Tensile strength psi (ultimate) (Min.)	Elongation in 2 inches (Min.)	Reduction of area (Min.)	Brinell hardness number (Min.)
Pounds	Pounds	Pounds	Percent	Percent	
29,000,000	45,000	90,000	35	50	240

3.4.2.2 External straightness. - Permissible variation in straightness throughout the length of the tube shall be not more than 0.010 inch, when tested as specified in 4.4.2.3.

3.4.2.3 Concentricity of tube. - The center of the bore shall not, at any point, deviate more than 0.005 inch from the center of the external surface of the main body tube at that point, when tested as specified in 4.4.2.4.

3.4.2.4 Concentricity of taper. - The center of the bore shall not, at any point, deviate more than 0.010 inch from the center of the external surface of the taper at that point, when tested as specified in 4.4.2.5. The eccentricity of the threads when used and external surface mating with the main body tube shall be not more than 0.003 inch, when tested as specified in 4.4.2.5.

3.4.2.5 Straightness of taper. - Permissible variation in straightness throughout the length of the taper shall be not more than 0.010 inch, when tested as specified in 4.4.2.6.

3.4.3 Magnetic permeability. - The magnetic permeability shall not exceed 3.00 (air equals 1.0).

3.4.4 Hydrostatic or air pressure. - The tubes and the tapers shall withstand a hydrostatic or air pressure of 1000 pounds per square inch (psi), without showing any weeping or leakage, when tested as specified in 4.5.1.

3.4.5 Symmetricity. - The end surfaces of the tube and the taper and all threads, grooves, and collars shall be symmetrical about their axes and in planes at right angles to this axis.

3.5 Finish. -

3.5.1 External surface. - The external surface of the tube shall be finished as specified on the applicable drawings. The finish shall be obtained by grinding, lapping or polishing, and shall be free from tool marks, flats, scars, and any other surface defects. The external surface of the taper, except the external surface of the shoulder at the large end of the taper, shall be sandblasted or otherwise roughened as specified on the applicable drawings to provide a surface suitable for painting. The external surface of the shoulder shall have the same outside diameter and shall be finished in the same manner as the external surface of the tube.

3.5.2 Internal surface. - The internal surface of the tube and taper shall be smooth and free from objectionable shoulders, cuts, or tool marks. The finish shall be as specified on the applicable drawings.

3.6 Marking. - The tubes and tapers shall be marked to indicate the matching units, if applicable, and for identification as specified (see 6.1).

3.7 Workmanship. -

3.7.1 The tubes and tapers shall be of uniform quality and condition, free from seams, pipes, flaws, cracks, scale, fins, porosity, hard spots, sand, excessive nonmetallic inclusions and segregations, and any other defects which may effect the serviceability.

3.7.2 Tubes and tapers shall not be welded, plugged, plated, peened, coated, impregnated or repaired in any manner to obtain hermetical tightness.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all the inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records for each tube and each taper shall include the following:

(a) Melt number.
(b) Contract or order number.
(c) Item number.
(d) All measurements of finished dimensions.
(e) Complete physical characteristics.
(f) Chemical composition.

4.2 Lot. - Tubes and tapers shall be considered individually for acceptance inspection; consequently each tube and taper shall be a lot for chemical analysis, physical properties tests, microscopic examination and other inspection.

4.3 Sampling for tests of forgings. - Each tube and each taper shall have extra material, in the forging, for the physical properties test specimen, magnetic permeability measurement piece, and microscopic examination specimen. There shall be enough extra material to permit these three tests on each end of each forging. The physical properties and permeability specimens shall be parallel to and in continuation of the metal forming the finished section of the tube or taper, in order that the structure and properties of both may be the same. The specimen for microscopic examination shall be from a transverse section of the forging. Millings or drillings for individual chemical analyses shall be taken from the physical properties test specimens from each end of each forging.

4.4 Examination. -

4.4.1 Visual and dimensional examination. - The entire surface of each tube and each taper shall be examined to determine compliance with 3.4.5, 3.5 and 3.7. All dimensions shall be measured to assure conformance with the applicable drawings.

4.4.2 Dimensions. - Measurements of the dimensions shall be taken as specified in 4.4.2.1 through 4.4.2.3.

4.4.2.1 External diameter. - Micrometer measurements of 2 diameters 90 degrees apart shall be made and recorded at intervals of 6 inches throughout the length of the tube and at such points on the taper as deemed necessary.

4.4.2.2 Internal diameter. - Star gage measurements shall be made and recorded at intervals of 6 inches throughout the length of the tube and the taper.

4.4.2.3 External straightness of tube. - Each tube shall be supported horizontally in two V-blocks or pairs of rollers. These supports shall, in general, be placed near the ends, clear of the threads, grooves and counterbores, but may be located and shifted at the discretion of the inspector. The tube shall be slowly rotated by hand and a registering or indicating micrometer applied to the side of the tube at one end of the horizontal diameter. These micrometer readings shall be taken every 12 inches throughout the length of the tube. The total difference in reading of the micrometer for a complete revolution shall be not more than 0.020 inch.

4.4.2.4 Concentricity of bore of tube. - Each tube shall be supported as specified in 4.4.2.3. The tube shall be slowly rotated by hand and a registering or indicating micrometer applied to the bore surface of the tube at one end of the horizontal internal diameter. Micrometer readings shall be taken at points about 10 inches from each end, plus one, approximately 3 inches from the upper end of the tube directly over the supports. Similar micrometer readings shall be taken at intervals of 2 feet throughout the length of the tube. The total difference in reading of the micrometer for a complete revolution shall be not more than 0.010 inch.

4.4.2.5 Concentricity of taper. - Each taper shall be supported horizontally by its minimum external diameter and by the external spigot surface which mates with the main body tube, in two V-blocks or pairs of rollers. These supports shall be placed near the ends, clear of the threads and non-cylindrical surface. The taper shall be slowly rotated by hand and a registering or indicating micrometer applied to an internal cylindrical surface at a point within 6 inches of the head end, at the end of a horizontal diameter. The total difference in reading of the micrometer for a complete revolution shall be not more than 0.020 inch. The micrometer shall also be applied to the internal tapered surface at a point within 6 inches of the lower end of the taper, at the end of a horizontal diameter. The total difference in reading of the micrometer for a complete revolution shall be not more than 0.010 inch. The micrometer shall also be applied to the external cylindrical surface which has the same nominal diameter and finish as the main body tube. The total difference in reading of the micrometer for a complete revolution shall be not more than 0.006 inch. To check for concentricity of the external thread, when used, an indicating micrometer shall be placed on one end of a horizontal diameter of an intermediate ring gage or against a wire contacting the thread at its pitch diameter. The total difference in micrometer reading for one revolution of the taper shall not exceed 0.004 inch.

4.4.2.6 Straightness of taper. - Each taper shall be fully screwed into the corresponding tube or held firmly in the tube with the tube and taper shoulders together, and when no threads are specified. The tube shall be supported as specified in 4.4.2.3 and shall be slowly rotated by hand. A registering or indicating micrometer shall be applied to a

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cylindrical surface at a point within 6 inches of the head end of the taper at the end of a horizontal diameter. The total difference in reading of the micrometer shall be not more than 0.020 inch.

4.4.3 When matched pairs of threaded tubes and tapers are specified (see 6.1) each taper shall be fully screwed into the tube to which it is to be matched and removed prior to acceptance.

4.5 Tests.-

4.5.1 Pressure tests.- Each tube and each taper, after being completely finished, shall be subject to either of the tests specified in 4.5.1.1 and 4.5.1.2.

4.5.1.1 Air pressure.- The ends of the tube and the taper shall be plugged and each completely immersed in a tank of still water and subjected to an internal air pressure of 1000 psi for 30 minutes. Any observed leakage of air shall be considered a failure and the tube or taper shall be rejected.

4.5.1.2 Hydrostatic pressure.- The ends of the tube and taper shall be plugged and each subjected to an internal hydrostatic pressure of 1000 psi for 30 minutes. Any weeping or other leakage of the test water shall be considered a failure and the tube or taper shall be rejected.

4.5.2 Physical properties.- Tests shall be performed on at least 2 test specimens from each forging (see 4.3) as specified in 4.5.2.1 through 4.5.2.5.

4.5.2.1 Modulus of elasticity.- The modulus of elasticity is the ratio of the increment of unit stress to increment of unit deformation within the elastic limit. These values shall be determined by use of an extensometer as follows: The test specimen shall be initially loaded to 15,000 psi and the deformation recorded. The load shall be increased in steps of 2500 psi until a load of 40,000 psi has been attained. Deformations shall be measured and recorded for each load. The unit load shall be plotted against the unit deformation for determination of the modulus of elasticity.

4.5.2.2 Proof stress.- The proof stress is that stress which produces a permanent elongation of 0.0002 inch in 2 inches after release of the load. Determination of this value shall be a continuation of the test to determine the modulus of elasticity (see 4.5.2.1). After a load of 40,000 psi inch has been reached, additional loads shall be applied in increments of 2500 psi until the permanent elongation of 0.0002 inch in 2 inches has been exceeded as indication by extensometer readings after release of the load. The last reading taken prior to the point where the permanent elongation of 0.0002 inch in 2 inches is exceeded shall be recorded as the proof load from which the proof stress shall be calculated.

4.5.2.3 Tensile strength.- The tensile strength is the maximum load sustained by the test specimen prior to rupture divided by the original area at that point. This maximum load shall be determined by removing the extensometer from the test specimen used in 4.5.2.1 and 4.5.2.2 and gradually increasing the load until rupture takes place. The maximum load sustained prior to rupture shall be used for determination of the tensile strength.

4.5.2.4 Elongation and reduction of area.- After rupture of the test specimen used in 4.5.2.3 the elongation and reduction of area shall be checked for conformance with 3.1.2.

4.5.2.5 Brinell hardness.- Tests shall be performed to determine conformance with the Brinell hardness specified in 3.1.2.

4.5.3 Microscopic examination.- When specified (see 6.1), the microscopic examination specimens shall be examined for conformance with 3.7.1.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and preservation.- Material shall be cleaned and dried by a process or procedure in accordance with specification MIL-P-116 which will accomplish thorough cleaning and drying without damage to the item. Critical functioning or close tolerance surfaces shall be cleaned to insure removal of all corrosion, soil, grease, residues and fingerprints, perspiration or other acid and alkali residues. Application of preservative compounds will not be required.

5.2 Packaging and packing.- Each tube or taper shall be packaged and packed for shipment in Government furnished containers, stock numbers S8115-792-3662 or S8115-691-1458, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Title and number of applicable drawings (see 3.2 and 3.4.1).
(c) Diemensions of tubes and tapers (see 3.4.1).

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(d) Marking of tubes and tapers (see 3.6).
(e) When matched pairs of tubes and tapers (see 4.4.3).
(f) When microscopic examination should be performed (see 4.5.3).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Instructions

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specification and Standardization Branch, Washington 25, D. C.

Specification

Organization	City	State

Contract No.

Quantity of Items Procured	Dollar Amount $

Material procured under a direct Government contract ☐ or a subcontract ☐

1. Has any part of the specification created problems or required interpretation in procurement?
   a. Give paragraph number and wording.

   b. Recommendations for correcting the deficiencies.

2. Comment on any specification requirement considered too rigid.

3. Is the specification restrictive? ☐ Yes ☐ No
   If the answer is "Yes", in what way?

4. Remarks (Attach any pertinent data which may be of use in improving this specification.)
Place this form and papers in an envelope and send to the Bureau.

Submitted by (Print name and activity)	Date

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